One gauge for every record IAM reads

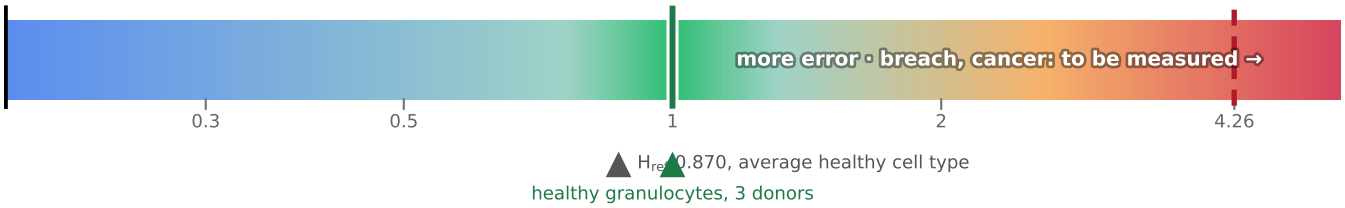
1 = healthy, in the middle · H_{min} = the floor where thermal kicks win; nothing reads below it · above 1 = more error

Cell · neutrophil · IAM-A

held error at each CpG, read on single molecules

H_{min} 1×10⁻⁷, off the left edge · (thermal kicks win against one ATP)

surface full 4.26

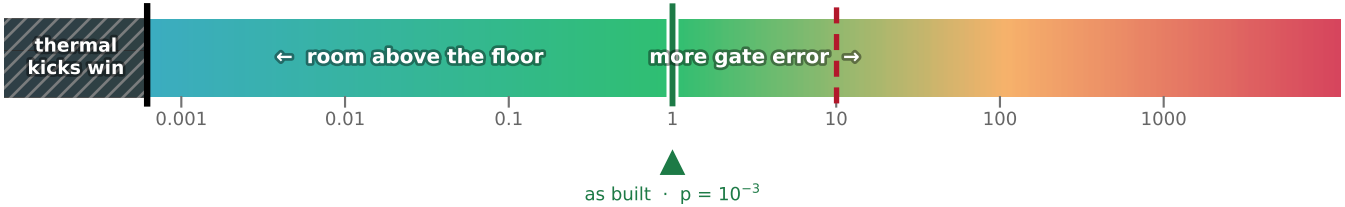


Qubit · transmon reading · two-qubit gate

ε = −ln(1 − p); 5 GHz at its own 35 mK, T1 68 µs, 40 ns gate, p = 10⁻³

H_{min} 6.2e-4 · thermal floor

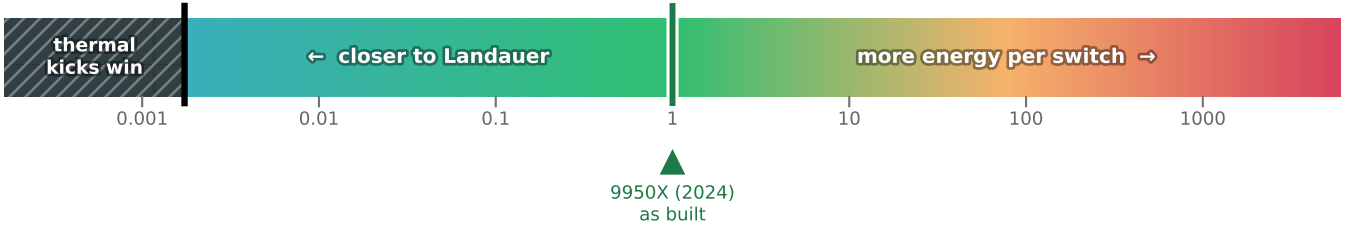
error correction fails · p = 1 %



Chip · AMD Ryzen 9 9950X · switching energy

E = TDP / (N f); the chip as built against k_B T_j ln 2

H_{min} 0.0017 · Landauer floor k_BT ln 2



A = reading ÷ the same system's healthy reading (log scale, each gauge centred on 1)